

PROJECT SPECIFICATION v1.0

Pre-Owned Vehicle Marketplace

Full Product Specification

Connecting customers with pre-owned vehicle showrooms

GEOGRAPHY	Single State to PAN India
VEHICLES	Bikes - Scooters - Cars - Autos - Commercial
PLATFORMS	Flutter Mobile + Next.js Web
VERSION	1.0 - June 2026



Table of Contents

1	Project Overview
2	User Roles
3	Tech Stack
4	Feature Specification by Role
5	Subscription Model
6	Verification System
7	Vehicle Data Handling
8	Chat & Communication
9	RC Transfer Tracker
10	Notifications
11	Analytics Dashboards
12	Feature Roadmap
13	System Architecture
14	Key Business Rules

A platform connecting customers with pre-owned car and bike showrooms, acting as a trusted digital middleman.

This is a pre-owned car and bike marketplace where the platform acts as a trusted middleman between customers and registered showrooms/dealers. The platform facilitates discovery and communication — the actual financial transaction happens directly between buyer and seller.

Core Idea

- Shop owners get a digital storefront (like an Instagram profile) for their entire inventory.
- Customers can browse, filter, compare, and enquire about vehicles.
- Each vehicle has an Enquire button — not a Buy button. The actual sale happens offline.
- The platform does not handle the financial transaction.

Launch Geography	Target Vehicles	Business Model
Single state first → PAN India	Bikes · Scooters · Cars · Autos · Commercial	Subscription (■299–■599/month)

Customer	Browse, wishlist, compare, and enquire about vehicles.
Shop Owner	List vehicles, manage inventory, chat with customers, track sales.
Association Admin	District-level partner account with limited analytics access.
Super Admin	Founder/founding team — full platform control.

Component	Technology	Purpose
Mobile App	Flutter	Single codebase for Android and iOS.
Web App	Next.js (React)	SSR for SEO on vehicle listing and shop profile pages.
Backend API	Node.js + Express	Single REST API shared between Flutter and Next.js.
Database	PostgreSQL	Primary relational database for all structured data.
Real-Time	Firebase	Real-time chat and push notifications via FCM.
Payments	Razorpay	In-app subscriptions via UPI, cards, and net banking.
Media Storage	Cloudinary	Vehicle photo uploads, storage, and CDN delivery.
Caching	Redis	Cache RTO API responses. Prevents redundant API calls.
Web Hosting	Vercel	Next.js SSR deployment with edge network.
API Hosting	Railway / AWS	Node.js backend and PostgreSQL hosting.

4.1 Shop Owner

Onboarding

- Register with name, phone number, shop name, location, and GST number.
- GST number validated via API before account is activated.
- Auto-approved after GST validation — 30-day free trial begins.
- Association Admin notified in background to review shop genuinity.

Shop Profile Page

- Instagram-style grid displaying all listed vehicles.
- Unique shareable URL per shop (e.g. platform.com/shop/shop-name).
- Guest view: photos visible, price and specs blurred until login.
- Includes shop name, location, verified badge, rating, and listing count.

Vehicle Listing

- Shop owner enters only the vehicle registration number.
- All details fetched automatically via RTO/vehicle API.
- Shop owner adds: asking price (mandatory), photos, and notes.

Inventory Management

- View all listed vehicles in grid or list view.
- Mark vehicle as Sold — triggers sale confirmation flow.
- Edit price or details at any time. Archive or delete listings.
- View all vehicle insurance expiry dates in one place.

Test Drive Booking

- Enable per listing with mode: At Shop or Home Delivery.
- Customer selects date/time. Shop owner approves or proposes alternative.
- Day-before reminder sent to both parties.

Sale Confirmation & Verified Badge

- Both shop owner and customer must confirm each sale.
- Verified Badge awarded automatically after 5 confirmed sales.
- RC Transfer Tracker automatically created on sale confirmation.

Shop Analytics Dashboard

- Total enquiries (all time and this month), enquiries per vehicle.
- Most viewed vehicles, wishlist saves per vehicle.
- Test drives booked vs completed, profile views from shareable URL.

4.2 Customer

Onboarding & Browsing

- Register with name, phone, and location.

- Guest browsing: photos and names visible, prices and specs blurred.
- Login required to see full details or send an enquiry.

Search & Filters

- Filter by vehicle type, location/district, price range, year, fuel type, brand.
- Search for shops by name or location.
- New vehicle listings near the customer's location trigger push notifications.

Vehicle Detail Page

- Full RTO API data: CC, BHP, fuel, colour, insurance expiry, year.
- EMI Calculator: vehicle price pre-filled, customer sets down payment, tenure, interest rate.
- Enquire button opens the chat thread with the shop owner.

Wishlist & Comparison

- Save any vehicle to a personal wishlist. Price drop alerts when price reduces.
- Compare up to 3 vehicles side-by-side: CC, BHP, insurance, colour, price, year.

Reviews

- Any customer who has sent an enquiry to a shop can leave a public review.
- Star rating + written text. Visible on the shop's profile page.

4.3 Association Admin

- One Association Admin account per district — partnered with the platform.
- Dashboard: total shops, active subscriptions, confirmed sales count (no revenue data).
- Receives notification when a new shop registers in their district.
- Can flag suspicious shops to the Super Admin.
- Cannot access vehicle prices, subscription amounts, or any financial data.

4.4 Super Admin

- Platform-wide dashboard: shops, listings, enquiries, subscriptions, sales.
- Manage all shops: view, suspend, activate, extend subscriptions manually.
- Review and resolve flagged shops and listings from customers and association admins.
- Create and manage Association Admin accounts per district.
- Push banners and announcements to all shop owners or all customers.
- View GST verification log and Razorpay payment history.

Feature	Basic — ■299/month	Premium — ■599/month
Vehicle listings	Unlimited	Unlimited
Photos per vehicle	5 photos	15 photos
Search placement	Standard	Top of results
Push to nearby customers	No	Yes — on new listing
Featured badge on listings	No	Yes
Text chat with customers	Yes	Yes
Phone number reveal	Yes	Yes
Free trial	30 days	30 days

Expiry Behaviour

- Listings remain visible with an 'Inactive Shop' label.
- Enquire button is disabled on all listings.
- Existing chat threads remain accessible.
- Renewal reminders sent 7 days before expiry and immediately after expiry.

GST Verification

- Required at shop registration. Validated via GST API before account activation.
- If GST fails, the shop owner is prompted to re-enter or contact support.
- GST verification log is visible to Super Admin.

Association Admin Background Review

- After GST passes and account is auto-approved, Association Admin is notified.
- Shop is live immediately — this is a background review, not a gate.
- If suspicious, Association Admin can flag to Super Admin.

Verified Badge

Automatically awarded after 5 confirmed sales where both parties confirmed each sale. Cannot be purchased.

How Vehicle Listing Works

Step 1: Shop owner enters registration number only.

Step 2: Platform calls RTO/vehicle API to fetch all details automatically.

Step 3: API response cached in Redis (TTL 30 days) and persisted to database.

Step 4: Shop owner adds: asking price (mandatory), photos, and notes.

Step 5: Listing goes live on the shop profile and search results.

Fields Fetched from API

Field	
Make, model, and variant	
Year of manufacture	
Engine CC and BHP	
Fuel type and colour	
Insurance expiry date	
RC owner name and RC status	
Chassis number and engine number	

API Rule: Called once at listing creation, then automatically once per month for refresh. Never on every page view.

- Text-only chat between customer and shop owner, powered by Firebase Firestore.
- Chat thread created when a customer clicks Enquire on any vehicle.
- Shop owner's phone number is revealed inside the chat thread.
- Customer can then call the shop owner directly outside the app.
- Shop owner receives a push notification for every new enquiry.
- v1: Shop owners can tag enquiries as New / In discussion / Test drive scheduled / Sold.

Triggered automatically after both parties confirm a sale. Tracks the RC ownership transfer process.

Stage	Description	Who Updates
1 — Documents collected	All required documents collected from the buyer	Shop owner
2 — Form 29/30 submitted	Transfer application submitted at the RTO office	Shop owner
3 — Hypothecation cleared	Bank's hypothecation removed (if vehicle had a loan)	Shop owner
4 — RC copy received	New RC with updated ownership details received	Shop owner
5 — Transfer complete	Full ownership transfer completed	Shop owner

If a stage has not changed in 7 days, both the shop owner and customer receive an automated reminder notification.

Notification	Recipient	Trigger
New enquiry alert	Shop owner	Customer sends first message
Price drop alert	Customer	Wishlisted vehicle price reduced
Test drive reminder	Both parties	Day before scheduled test drive
Insurance expiry	Shop owner	30 days before vehicle insurance expires
Subscription renewal	Shop owner	7 days before subscription expires
New vehicle nearby	Customer	New listing in customer's district
RC transfer stuck	Both parties	Transfer stage unchanged for 7 days
Verified badge awarded	Shop owner	Shop reaches 5 confirmed sales
New shop registered	Association Admin	New shop signs up in their district
Shop genuinity flag	Super Admin	Association Admin flags a shop

All push notifications delivered via Firebase Cloud Messaging (FCM).

Shop Owner Dashboard

Metric	Description
Total enquiries	All-time and this month
Enquiries per vehicle	Which listings are getting the most interest
Most viewed vehicles	Top listings by page views
Test drives booked	Total booked vs completed
Wishlist saves	How many customers saved each vehicle
Profile views	Views from the shareable shop URL
Subscription status	Current plan and days remaining

Association Admin Dashboard

Metric	Description
Shops in district	Total registered shops
Active subscriptions	Shops with active plans
Confirmed sales	Total sale count in district (no amounts)
New shop alerts	Shops pending genuinity review

Super Admin Dashboard

Metric	Description
Active shops	Subscribed shop count
Shops in trial	Free trial shops
Total vehicles listed	Platform-wide inventory count
Total enquiries	Platform-wide enquiry count
Confirmed sales	Total sales across all shops
Subscription health	Active vs expired subscriptions
Flagged shops	Shops under review
Verified badges given	Count of verified shops

MVP — Months 1 to 3

Core loop: list → browse → enquire

- Auth & onboarding — GST verification, 30-day trial, role-based accounts
- Vehicle listing — registration number → auto-fetch via RTO API
- Shop profile page — shareable URL, blurred guest view
- Inventory management — mark sold, edit price, manage status
- Browse & search — filters by type, location, price
- Enquiry & chat — text chat + phone number reveal
- Wishlist
- Subscription & payments — ₹299/₹599 via Razorpay, inactive label on expiry
- Super admin panel — shops, listings, subscriptions, flagging
- Core notifications — new enquiry, subscription renewal, insurance expiry

v1 — Months 4 to 7

Trust, transactions & discovery

- Premium boost — top of search, push to nearby users, more photos
- Shop analytics dashboard
- Test drive booking — at-shop or home delivery, day-before reminder
- Sale confirmation flow — both parties confirm
- RC transfer tracker — 5-stage pipeline, 7-day stuck alert
- Verified badge — auto-awarded after 5 confirmed sales
- Vehicle comparison — up to 3 vehicles
- EMI calculator — on vehicle detail page
- Public reviews — post-enquiry, visible on shop profile
- Association admin dashboard — district analytics, genuinity notifications
- Full notifications — price drop, test drive reminder, new vehicle nearby, RC stuck

v2 — Months 8 to 12

Scale to PAN India + growth features

- Service history upload — shop attaches past service records to vehicles
- Enquiry inbox status tags — new → discussing → test drive → sold
- Referral program — invite shops, earn subscription credits
- Similar vehicle recommendations — on vehicle detail page
- Vehicle condition grading — standardised A/B/C grade on listing cards
- Multilingual support — Tamil, Hindi, Telugu, Kannada
- Monthly downloadable reports for Association Admins

- Banner/announcement system for Super Admin
- PAN India expansion — multi-state association admins, geo-based boosting
- Automatic monthly vehicle API refresh — scheduled background job

Layer	Detail
Client Layer	Flutter Mobile (iOS + Android) + Next.js Web App (SSR for SEO)
API Layer	Node.js + Express REST API — shared backend for both clients
Database	PostgreSQL (primary) with 13 domain schemas
Real-Time	Firebase Firestore for chat · FCM for push notifications
Cache	Redis — vehicle API cache, search results, session tokens
Media	Cloudinary — vehicle photos, CDN delivery, signed uploads
Payments	Razorpay — subscription orders, webhooks, UPI/card/netbanking
External APIs	RTO/Vehicle API (reg number lookup) · GST API (shop verification)
Hosting	Vercel (Next.js) · Railway/AWS (API + DB) · Upstash (Redis)

Scheduled Background Jobs

Job	Schedule	Action
Monthly vehicle API refresh	1st of every month	Re-fetch RTO data for all active listings
Subscription expiry check	Daily at 8 AM	Flag expired subscriptions, send renewal reminders
RC transfer stuck check	Daily at 10 AM	Find transfers unchanged for 7+ days, send reminders
Insurance expiry reminders	Daily at 9 AM	Find vehicles expiring within 30 days, notify shop owners

Rule	Detail
Free trial	30 days full access. Subscription required after trial ends.
Listing limit	No limit on either plan. Differentiated by boost features.
Subscription expiry	Listings visible, enquiry button disabled, 'Inactive Shop' label shown.
Sale confirmation	Both shop owner and customer must confirm. Only then does the sale count increase.
Verified badge	Auto-awarded at 5 confirmed sales. Cannot be purchased.
Vehicle API calls	Called once at listing creation, then once per month. Never on every page view.
Reviews	Only customers who have sent an enquiry to the shop can leave a review.
GST validation	Required to activate. Auto-approved after passing. Association Admin notified.
Guest browsing	Photos and names visible. Price and specs blurred. Login required to enquire.
Association Admin access	Can see shop count and sale count only. No revenue, price, or financial data.
Revenue model	Subscription only. Commission per sale may be added in a future version.
Geography	Single state at launch. PAN India in v2.
Languages	English at launch. Tamil, Hindi, Telugu, Kannada in v2.
Super admin role	Founder only. Not assignable to other staff.

Pre-Owned Vehicle Marketplace — Project Specification v1.0

June 2026 · Confidential — For internal and development team use only